

STACK PATTERNS – COMPLETE INTERVIEW SHEET

All Stack Patterns (Quick Index)

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Monotonic Stack (Next Greater / Smaller)

When to use

- Next / Previous greater or smaller element
- First element on left/right satisfying condition

Questions

- Next Greater Element I
<https://leetcode.com/problems/next-greater-element-i/>
Companies: Amazon, Microsoft, Google
 - Next Greater Element II
<https://leetcode.com/problems/next-greater-element-ii/>
Companies: Google, Amazon
 - Daily Temperatures
<https://leetcode.com/problems/daily-temperatures/>
Companies: Amazon, Meta, Microsoft
 - Next Smaller Element (GFG)
<https://practice.geeksforgeeks.org/problems/next-smaller-element/>
Companies: Amazon, Flipkart
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Stock Span Pattern

When to use

- Count consecutive previous elements

- Span or streak based logic

Questions

- Online Stock Span
<https://leetcode.com/problems/online-stock-span/>
Companies: Amazon, Microsoft
 - Stock Span Problem (GFG)
<https://practice.geeksforgeeks.org/problems/stock-span-problem/>
Companies: Flipkart, Infosys
 - Number of Visible People in a Queue
<https://leetcode.com/problems/number-of-visible-people-in-a-queue/>
Companies: Google
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Histogram / Area Pattern

When to use

- Maximum area / rectangle problems
- Expand left and right boundaries

Questions

- Largest Rectangle in Histogram
<https://leetcode.com/problems/largest-rectangle-in-histogram/>
Companies: Google, Amazon, Uber
 - Maximal Rectangle
<https://leetcode.com/problems/maximal-rectangle/>
Companies: Amazon, Microsoft
 - Trapping Rain Water
<https://leetcode.com/problems/trapping-rain-water/>
Companies: Google, Meta
 - Sum of Subarray Minimums
<https://leetcode.com/problems/sum-of-subarray-minimums/>
Companies: Amazon
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Parentheses & Bracket Validation

When to use

- Balanced or nested structures

Questions

- Valid Parentheses
<https://leetcode.com/problems/valid-parentheses/>
Companies: Amazon, Microsoft, PayPal
 - Longest Valid Parentheses
<https://leetcode.com/problems/longest-valid-parentheses/>
Companies: Google
 - Minimum Add to Make Parentheses Valid
<https://leetcode.com/problems/minimum-add-to-make-parentheses-valid/>
Companies: Adobe
 - Remove Invalid Parentheses
<https://leetcode.com/problems/remove-invalid-parentheses/>
Companies: Meta
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Expression Evaluation Stack

When to use

- Infix / Prefix / Postfix expressions
- Operator precedence

Questions

- Evaluate Reverse Polish Notation
<https://leetcode.com/problems/evaluate-reverse-polish-notation/>
Companies: Amazon, Oracle
 - Basic Calculator II
<https://leetcode.com/problems/basic-calculator-ii/>
Companies: Uber, Airbnb
 - Basic Calculator
<https://leetcode.com/problems/basic-calculator/>
Companies: Google
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Stack Design & Simulation

When to use

- Custom stack behavior
- Extra operations in $O(1)$

Questions

- Min Stack
<https://leetcode.com/problems/min-stack/>
Companies: Amazon, Google
 - Max Stack
<https://leetcode.com/problems/max-stack/>
Companies: LinkedIn
 - Implement Queue using Stacks
<https://leetcode.com/problems/implement-queue-using-stacks/>
Companies: Amazon, Infosys
 - Implement Stack using Queues
<https://leetcode.com/problems/implement-stack-using-queues/>
Companies: TCS
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Stack + Greedy (Remove / Optimize)

When to use

- Remove elements to get optimal result

Questions

- Remove K Digits
<https://leetcode.com/problems/remove-k-digits/>
Companies: Google, Amazon
 - Remove Duplicate Letters
<https://leetcode.com/problems/remove-duplicate-letters/>
Companies: Meta
 - Make The String Great
<https://leetcode.com/problems/make-the-string-great/>
Companies: Amazon
 - Smallest Subsequence of Distinct Characters
<https://leetcode.com/problems/smallest-subsequence-of-distinct-characters/>
Companies: Google
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Stack + String Processing

When to use

- Undo / backspace
- Pattern removal

Questions

- Backspace String Compare
<https://leetcode.com/problems/backspace-string-compare/>
Companies: Adobe
 - Simplify Path
<https://leetcode.com/problems/simplify-path/>
Companies: Airbnb
 - Decode String
<https://leetcode.com/problems/decode-string/>
Companies: Microsoft
 - Remove All Adjacent Duplicates II
<https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string-ii/>
Companies: Amazon
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Contribution / Subarray Stack

When to use

- Each element contributes as min/max

Questions

- Sum of Subarray Minimums
<https://leetcode.com/problems/sum-of-subarray-minimums/>
Companies: Amazon
 - Sum of Subarray Ranges
<https://leetcode.com/problems/sum-of-subarray-ranges/>
Companies: Google
 - Subarray Minimums (GFG)
<https://practice.geeksforgeeks.org/problems/sum-of-subarray-minimum/>
Companies: Flipkart
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Stack for DFS / Iterative Traversal

When to use

- Replace recursion
- Explicit stack control

Questions

- Binary Tree Inorder Traversal (Iterative)
<https://leetcode.com/problems/binary-tree-inorder-traversal/>
Companies: Amazon, Microsoft
 - DFS of Graph (Iterative – GFG)
<https://practice.geeksforgeeks.org/problems/depth-first-traversal-for-a-graph/>
Companies: Google
 - Clone Graph
<https://leetcode.com/problems/clone-graph/>
Companies: Meta
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How to Use This Sheet

- Patterns 1–4 → Core stack mastery
- Patterns 5–7 → FAANG depth
- All patterns → Interview-proof